

A Preliminary Study of World Customizability for Virtual Reality Co-Play

Cheryl Lao
cheryl.lao@uwaterloo.ca
University of Waterloo
Canada

Craig S. Kaplan
csk@uwaterloo.ca
University of Waterloo
Canada

Yanting Zhang
yanting.zhang@uwaterloo.ca
University of Waterloo
Canada

Morgan McGuire
morgan@roblox.com
Roblox Research
USA
University of Waterloo
Canada

Daniel Vogel
dvogel@uwaterloo.ca
University of Waterloo
Canada

Victor B. Zordan
vzbordan@roblox.com
Roblox Research
USA
Clemson University
USA

ABSTRACT

Cooperative play, or “co-play”, is the act of playing with others in a co-located setting, including co-location of avatars in virtual reality (VR). Customizability is the degree to which play artifacts like props can be changed to suit different needs, a factor of co-play which is easier to support in VR than in the real world. We present the results of a preliminary user study that explores how different levels of customization affect creativity in a two-person VR co-play setting. Using the Creativity Support Index, system logs, and observations, we found that increasing customizability of props used to improvise a story trended toward higher levels of perceived creativity. Our work introduces a new topic of investigation for social VR together with a study methodology and initial results to motivate further investigation.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing design and evaluation methods*; **Human computer interaction (HCI)**.

KEYWORDS

Virtual Reality, Co-Play, Social VR, Creativity

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1 INTRODUCTION

Partens [22] originally defined *cooperative play* as a situation where “the child plays in a group that is organized to attain some competitive goal, or of dramatizing situations of adult and group life, or of playing formal games”. Typically referred to as “co-play”, it has been studied in diverse contexts, such as between parents and children [26], with the elderly [10], and over live streams where presenters play video games together and spectators play in parallel [3].

Artifact selection and modification for a certain goal are key aspects of co-play. Ivory et al. [16] identified that certain types of toys, such as dress-up clothes and giant pillows, encourage higher levels of co-play. A child might select a doll as a character in an improvised story and customize it by changing its clothes. We refer to this kind of object or world modification as *customization*. The ability to customize or construct a real-world object, such as furniture, can increase its perceived value for the creator, known as “The IKEA Effect” [21]. In virtual play environments, customization can take the form of personalizing an avatar with different clothing or changing the colours of objects in the environment. Customization can also affect virtual behaviour. For example, people who modify their avatar appearance have increased self-affirmation and self-expression in a virtual world [17].

Customization is often studied in the context of co-located play in the real world, such as block play [29] and train construction [25]. However, virtual worlds can act as spaces where players can feel co-located despite being physically separated. VR is an immersive medium, which allows for interactions such as non-verbal communication [18], emotion sharing [13], and self-expression through virtual avatars [8]. VR worlds allow players to customize their environment without physical limitations and create virtual objects in situ. There is also work on co-creation platforms in VR where users can interact and collaborate to create virtual objects with more flexibility than in the real world [12]. Avatars, environments, and interactions can be easily changed, unlike objects in the physical world. We are excited about VR environments for co-play because they are not constrained by physical limitations in their appearance or interactions, which enables new interactions between users and their environment. This is especially relevant for “social VR”, experiences that foster interactions and shared activities between users in a virtual space [23].

While past work has explored co-play in the physical world [10, 26] or through online games [3], we explore co-play in VR. Our specific research question is: “Does increased world customizability make players more creative during VR co-play?”. To this end, we designed a within-subjects experiment where pairs of participants improvise stories using virtual props with three different levels of prop customizability: 1) *no* customization, where props could only be selected and used; 2) *limited* customization, where prop colours could also be changed; and 3) *broad* customization, where props may also be combined to form new objects. To measure creativity, we collect self-reported scores for the Creativity Support Index [5], system event logs of user actions, and observations from play sessions. We present results from a preliminary study with 14 teens and youths, a population selected for their exposure to multiplayer online experiences. Our work introduces a new topic of investigation for social VR, provides some promising early results, and contributes a methodology for an expanded study in the future. Results suggest that increasing customizability of objects in the play environment results in increased levels of creativity.

2 RELATED WORK

There is a large body of existing work on co-play in various forms; here we focus on co-play using multiplayer digital media. Co-play has been used as a digital therapy tool [15], a way to make online games more enjoyable and challenging [9], and as a tool to strengthen parent-child relationships [26]. Co-play interactions often involve ad-hoc roles determined by the players in the activity. For example, players may pretend to be different characters in a story and then swap roles [11]. Players can also engage in mentor-mentee [27] or parent-child dynamics [2], where one player may be teaching or guiding the other. Joint Media Engagement, or co-play with digital media, can support parents in connecting with their children through play [2]. Co-play provides scaffolding for children to learn in the presence of others. For example, Martinez et al. [19] observed co-play between a parent and her children in *Animal Crossing* where the children combined their knowledge of flowers and fish, facilitated by their parent. Past work on parent-child co-play found mixed results for enjoyment in co-creation, but observed empathy and feelings of social connectedness [11].

Virtual customizations are any changes that a user makes to a virtual environment or the objects in it. Often referenced in business contexts, the IKEA Effect describes the increase in valuation of self-made products [21]. Extended to virtual environments, assets that a user exerts effort to build or customize may be seen as more valuable to them. This increased pride in their work may improve the player’s experience in the virtual environment. Toys can serve as an extension of self for children and help them in embodying certain traits, characteristics or emotions [7]. The ability to customize virtual avatars has been shown to improve self-affirmation and self-expression in virtual worlds, leading to a more enjoyable experience [17].

Social VR facilitates live interactions between players who are co-located in a virtual reality space. Marcello et al. [1] found that virtual co-location can be just as effective for communication as playing VR games simultaneously in the same room. The range of emotional states in VR is also similar to that experienced by face-to-face groups

[20]. This virtual communication of emotion makes VR a suitable medium for performing improvised interactive dramas with others [28]. Multiplayer sandbox-style gaming environments offer players the ability to explore and display their creativity with others in a virtual world. Past works [4, 6] used *Minecraft*, a sandbox-style game, as a creative tool to study how virtual storytelling can be facilitated in a virtual environment using a shared building tool. As social VR grows in popularity, co-play between users who are virtually co-located will become increasingly common.

To our knowledge, no prior studies have explored how customizability affects creativity in VR co-play where players are immersed in a world without physical constraints.

3 VIRTUAL CO-PLAY ENVIRONMENT

We designed three separate Roblox¹ experiences for co-play. Participants were spawned on opposite ends of a virtual $1\text{m} \times 3\text{m}$ table in a featureless playground area within view of the props. Player avatars have R6 Roblox heads and cubes as hands that follow each VR controller. Players can scale themselves up and down using buttons on their controllers. The players grab props using the palm triggers and can move through the space using the joysticks. We developed our study environment in using *Luau* in Roblox Studio, selected for its support for multiplayer VR experiences. Assets were selected from the Roblox Toolbox, which is a freely available library of assets in Roblox Studio. These assets are user-generated and available to all creators on the platform, so selecting assets from this library emulates what the typical Roblox player would do during a game design session. Participants used Meta Quest 2 HMDs to access the experience on Roblox Player. Each participant had a physical area of approximately $2\text{m} \times 3\text{m}$ in which they could freely move without overlapping with the study area of their partner.

Please refer to the supplementary video for demonstrations of the conditions. The affordances in each of the levels are cumulative, meaning the features for conditions with higher customizability are a superset of the ones with lower levels of customizability. Three conditions are as follows:

No Customization (Fig. 1): Players are provided with a set of school supplies and container props along with the ability to scale them using a button press and duplicate them by tapping a duplication sphere in the scene. We do not consider duplication or scaling as customizations of props because duplication does not change any prop attributes and scaling is necessary to keep the props proportional in the case of a scaled player avatar.

Limited Customization (Fig. 2): Players can change the colours of props in the scene by tapping them on 9 colour-changing blocks in the scene behind the selection of props. Props can also be scaled and duplicated as they were in the condition with no customizability.

Broad Customization (Fig. 3): In addition to the capabilities of the other conditions, users also have the ability to attach props in the scene together. An attachment creator on the user’s virtual hand allows them to create small green attachments that connect two props when touched together (Fig. 4). By attaching multiple scene props together, the user is able to create composite props that better match their vision.

¹<https://www.roblox.com/home>



Figure 1: No customizability

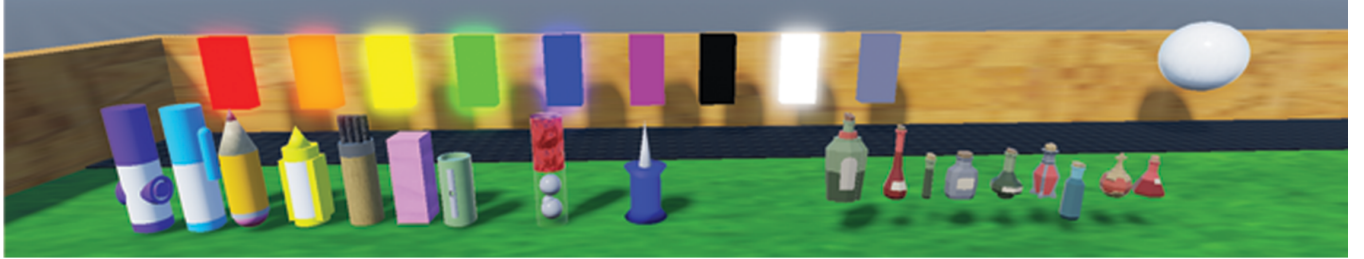


Figure 2: Limited customizability



Figure 3: Broad customizability

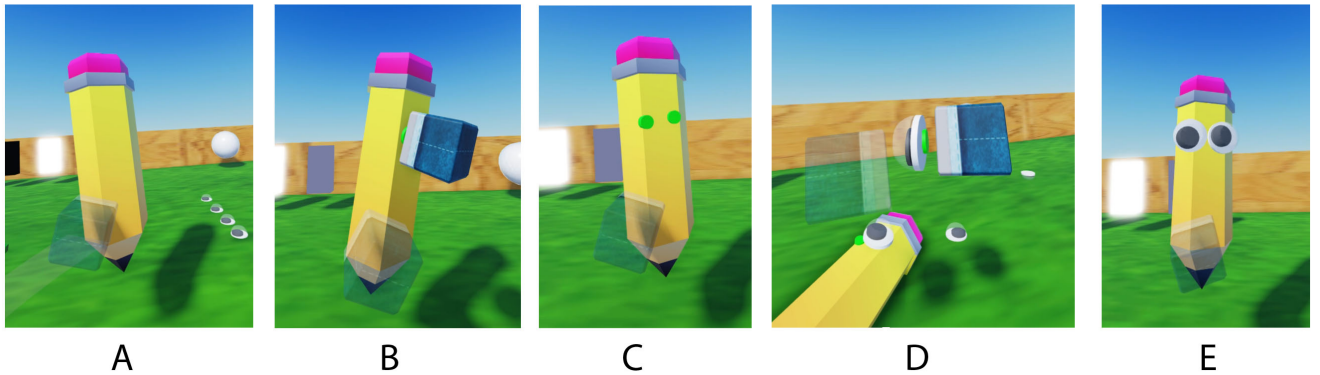


Figure 4: Example of assembling a custom prop in the broad customizability condition: (a) a base prop resembling a pencil is held one hand; (b) the other hand adds “attachment points” visualized as small green cylinders; (c) after two attachment points are added; (d) other props resembling cartoon eyes are attached to the base prop; (e) the final customized prop.

4 EXPERIMENT PROTOCOL

Our goal is to examine how different levels of customizability affect creativity in a co-play improvisational storytelling task. Pairs of younger participants cooperatively played together to tell stories using three different levels of customizability described above. In our study, pairs of participants engaged in three VR conditions with different virtual props and varying levels of customizability enabled. Participants created improvised scenes with the props

available to them. Teenagers were chosen as the primary demographic because co-play in virtual worlds is common in that age group. Prior research has shown that co-play between players who are familiar with each other differs from play with strangers [24]. This motivated our decision to recruit participant-pairs who were already familiar with one another.

4.1 Participants

We recruited 14 participants with a median age of 16 (mean=17.25, range=[15,24]), 6 identified as female, and 8 identified as male. Participants were recruited through advertisements at local high schools. They were asked to bring a friend to complete the study with them. This simulates how players of online social games usually play with friends with an established rapport. The study was approved by a research ethics board and each participant received a \$20 gift card. We express the participant IDs using their pair number with A or B differentiating between two participants in the pair.

4.2 Procedure

Setup (~5 minutes) Participants were given a brief introduction to the conditions of the experiment and VR hardware. They then adjusted the Quest 2 HMD for fit and interpupillary distance.

Creation (~3 minutes) Each participant selected an emotion (anger, sadness, disgust, fear, joy) to embody for the scene, without telling their partner. They then selected and customized props to be used in the improvisational scene.

Improvisation (~5 minutes) The experimenter provided a prompt from a list of improv acting prompts, randomly selected without replacement, to encourage their creative process. There were 6 possible prompts: *Two friends are auditioning for a talent competition*; *Two friends realize that they're on the wrong plane as it takes off*; *Two friends think they've just seen a ghost*; *Two friends are exploring a new planet*; *Two friends are at an amusement park on a rainy day*. These prompts were randomly assigned in each condition to reduce the bias that may arise from running all participants through the same scenarios in the same conditions.

Questionnaire/Break (~10 minutes) After each condition, participants filled out a questionnaire about creativity. This time also served as a break from the VR environment between conditions.

4.3 Design

This is a within-subjects design with one independent variable: CUSTOMIZABILITY with 3 levels (NONE, LIMITED, BROAD). The levels were assigned in a random order. Cherry and Latulipe's [5] Creativity Support Index (CSI) questionnaire provides six subjective measures that capture perceived creativity: *Collaboration*, *Enjoyment*, *Exploration*, *Expressiveness*, *Immersion*, and *Results Worth Effort*. It is similar to NASA TLX [14], but specifically designed to examine tools and tasks that support creativity. Measures computed from the logs (during customization and improvisation) were: *Hold Ratio*, the percentage of time that a prop is held over the course of a condition; *Customization Rate*, the number of customization actions (colour changes and attachments created) per minute of the condition; *Attachment Count*, the number of attachments created in the BROAD condition per minute; and *Grab Rate*, the number of times the user picks up a prop per minute. These objective logged measures help support our understanding for some CSI dimensions and may suggest trends in how participants adopted different strategies and behaviours for different customization conditions.

5 RESULTS

Measures and observations are organized by the CSI factors following Cherry and Latulipe's recommendation to report scores, counts,

and weighted scores. Each factor has a score out of 20, which is the sum of two 0-10 paired rating questions (Fig. 5). Participants were also asked to choose which factor was more important to them in forced choice comparisons between all choose two combinations (for a total of 15 comparisons). The count is the number of times each factor was chosen over another in that comparison (Fig. 6). Weighted scores for each creativity factor range from 0 to 100, where higher numbers indicate more creativity support. These are the product of count and score summed over all factors (Fig. 7). Table 1 provides all CSI results. We also follow Cherry and Latulipe's example of discussing means and trends (without formal statistical tests) with supporting observations. In the context of this preliminary study, this more exploratory analysis seems additionally appropriate.

5.1 Collaboration

Collaboration was encouraged throughout the study, but on the whole, pairs of participants began the exploration independently before showing their partners props that they discovered or created. Weighted scores suggest that as the level of customizability increases, so does the perceived level of support for collaboration (NONE=32.3, LIMITED= 36.6, BROAD=46.2). During the initial exploration of the BROAD condition, P1B said "Here you go, this is for you" and handed P1A a prop that they had selected. This suggests that participants are not only collaborating during storytelling, but also in the selection and customization of props for their partners.

Ownership: Participants expressed feelings of ownership over props that they selected, even without customization. In the NONE condition, P4B said "That's my pencil!" when P4A tried to pick up a pencil that they had previously used. We infer that players feel a sense of ownership over props after simply selecting them from the available props.

Recognizing the need for help: As a result of co-location in the virtual environment, participants could see their partners' actions. When P2A recognized that P2B was struggling to move a large table in the space, P2A attempted to help them to coordinate and move the table, even without an explicit request.

5.2 Enjoyment

Higher weighted scores for Enjoyment were reported in higher levels of customizability (NONE=45.4, LIMITED=52.1, BROAD=53.7). Enjoyment was the most heavily weighted factor across conditions (Fig. 6) with the counts very similar across conditions. This suggests that participants care about enjoying their co-play experience, regardless of the level of customization. P1A said "I love this island, it's so cool!" (referring to the play space as an island) in the BROAD condition, even before trying any of the customization features.

Holding Props: The average *Hold Ratio* was 21.5% for NONE, 26.6% for LIMITED, and 27.5% for BROAD, which suggests that increased customizability leads to higher levels of engagement with the virtual environment. The *Grab Rates* were 4.7, 4.17, and 4.9, respectively, which tells us that participants actually grabbed the most props in the NONE condition but did not engage with each prop for very long.

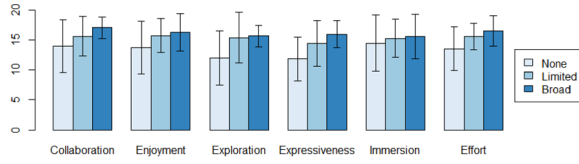


Figure 5: Mean CSI Scores with standard deviations

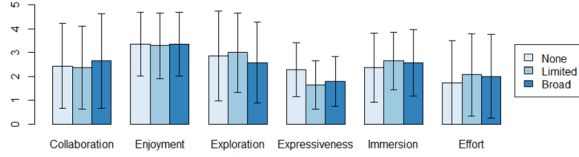


Figure 6: Mean CSI Counts with standard deviations

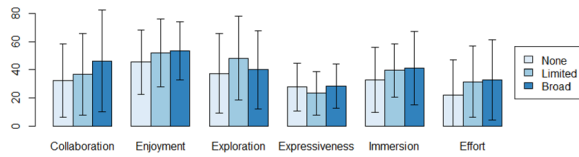


Figure 7: Mean Weighted CSI Scores with standard deviations

5.3 Exploration

The mean CSI scores show that the level of exploration increases with customizability (NONE=12, LIMITED=15.4, BROAD=15.7). However, with weighting from the counts, the LIMITED condition had slightly higher scores for exploration (NONE=37.5, LIMITED=48.2, BROAD=40.1). This may be a result of how the BROAD condition enables creation of composite props, which require more planning than colour changes.

5.4 Expressiveness

Weighted scores for Expressiveness were higher for BROAD and NONE conditions compared to LIMITED (NONE=27.7, LIMITED=23.3, BROAD=28.4). Surprisingly, counts for this factor were highest in the condition with no customizability (NONE=2.3, LIMITED=1.6, BROAD=1.8), in contrast with the trend of scores increasing with customizability. The contrast between the lack of creative options in that particular scene and the scene with more customizability may have prompted the participants to realize the importance of customizability for expressiveness.

Customizations: The *Customization Rate* was much higher in the LIMITED condition than the BROAD condition (LIMITED=2.5 actions/minute, BROAD=0.83 actions/minute). Participants were unable to change colours or create the attachments during the NONE condition. The difference in attribute customization frequency may be caused by participants spending more time planning the placement of the attachments in the BROAD condition. This suggests that participants preferred building composite props over customizing attributes on existing props. Alternatively, they may have spent more time building simply because it was a more complex task.

Imagination: Although limitations on customizability limited the participants' ability to create exactly the props that they envisioned, the constraints made participants more creative in assigning labels to props. For example, P7A and P7B pretended that a pizza

was a roller coaster (despite the lack of resemblance) and P7A pretended to "lock down" the roller coaster using a notebook and attachment. The locking mechanism was enabled by the attachments in the BROAD condition, but assigning a different meaning to the pizza could have occurred in any condition.

5.5 Immersion

Immersion scores tended to increase with customizability, but the differences were not as pronounced as with the other factors (NONE=32.7, LIMITED=39.6, BROAD=41.2). Players seemed to be fully immersed in the scene and embodied their characters even in the NONE condition. When P5B saw their partner scaling up their avatar, they said "You can be taller than me in [this] imaginary world".

Proximity: Our system did not use spatial audio, so the participants could communicate regardless of virtual distance. As a result, participants would navigate away in virtual space but continue collaborate through talking.

Assigning Meaning: While lower levels of customizability made harder to create props that suited the participants' mental image of the prop, it did encourage them to stretch their imaginations and assign meanings to props that did not bear a physical resemblance to the intended purpose. In the NONE condition, P6B pretended to use eraser as a phone. Their partner, P6A, later pretended to make a call on the same prop. The eraser was not explicitly reassigned to be a phone, but P6B understood the new meaning through P6A's interactions.

5.6 Results Worth Effort

Results Worth Effort describes the participant's feeling that their creation is worth the effort that they exerted to create it. There is an increase in the weighted scores for Results Worth Effort as the level of customizability increased (NONE=21.9, LIMITED=31.6, BROAD=32.9). While the effort needed to create a prop was higher in the BROAD condition, participants were happier with their props than the NONE condition where they were simply selecting them.

6 DISCUSSION

We found that perceived creativity levels increased with the level of customizability. This was supported by logs and observations from play sessions. Overall, these results suggest our conditions and protocol were successful for an initial study. However, there are several areas for exploration outside the scope of this work that we describe below.

Creativity Factors: Creativity is composed of many different factors that can be associated with user behaviours. However, the interpretation of behavioural metrics is difficult. For example, if a user picks up many props in the scene, are they more engaged because they are exploring different props or simply unsatisfied with the props? Future studies should allow participants to reflect on their experiences and give free-form explanations for their behaviour. The CSI is an index that allows users to rank their own perceptions across creativity factors, but it should be combined with other observations and free-form responses from the participants to form a holistic view of creativity support. While we did perform more formal analysis of CSI measures using an ANOVA, there was no significant difference. This was likely due to the lack of statistical

Table 1: Mean ($\bar{x}$) CSI scores, counts, and weighted CSI Scores with standard deviations (σ). Rows represent the different CSI factors: Collaboration, Enjoyment, Exploration, Expressiveness, Immersion, and Effort

	Scores						Counts						Weighted Scores					
	None		Limited		Broad		None		Limited		Broad		None		Limited		Broad	
	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ
Collaboration	14	4.5	15.6	3.3	17.1	1.8	2.4	1.8	2.4	1.7	2.6	2	32.3	26.1	36.6	29.1	46.2	36.3
Enjoyment	13.8	4.4	15.8	2.8	16.4	3.1	3.4	1.3	3.3	1.4	3.4	1.3	45.4	22.8	52.1	24.3	53.7	20.7
Exploration	12	4.6	15.4	4.3	15.7	1.8	2.9	1.9	3	1.7	2.6	1.7	37.5	28.3	48.2	29.8	40.1	27.8
Expressiveness	11.9	3.7	14.5	3.8	16	2.3	2.3	1.1	1.6	1	1.8	1.1	27.7	16.9	23.3	15.7	28.4	15.8
Immersion	14.5	4.7	15.3	3.2	15.6	3.7	2.4	1.4	2.6	1.2	2.6	1.4	32.7	23.3	39.6	19.1	41.2	26.1
Effort	13.6	3.7	15.6	2.2	16.5	2.6	1.7	1.8	2.1	1.7	2	1.8	21.9	25.5	31.6	25.3	32.9	28.7

power from a small sample size, or perhaps the subjective nature of the CSI. This may also be why Cherry and Latulipe [5] did not demonstrate CSI analysis with formal statistical methods.

Practice Sessions: Across different condition orderings, participants consistently enjoyed exploring the features and even the locomotion method in the VR scene in their first condition. Experimenters should consider having a preliminary exploration round to allow participants to familiarize themselves with the system. If participants are strangers, a non-VR practice session may help the participants feel more comfortable working with their partners.

Selection of Props: The selection of props may have affected how the participants felt in each stage of the study. The props in the BROAD condition were slightly more varied than in the other conditions to give the participants enough components to attach and create composite props. While this work explored a limited selection of props, future work may explore how the selection of available props influences player behaviour. The fidelity of the provided props (whether they are blocks or realistic renderings of props) may also affect the experience in each condition.

Demographics: The participants in our evaluation were teenagers who were enrolled in secondary school (with the exception of one pair who were aged 21 and 24). This demographic

was specifically chosen because collaborative online gameplay is very common in that age group and co-play with peers is still quite common. Recruiting younger children would have aligned better with existing co-play literature, but current VR health and safety recommendations state that users should be 12 or older. Future work may focus on different demographics or age groups, such as people with no online game experience or seniors.

7 CONCLUSION

In this work, we presented the results from a preliminary user study on the effects of customizability on co-play in VR. We found that customizability is associated with higher levels of perceived creativity support from the tool, but there is much work to be done to determine exactly how customizability impacts a user's experience. As social interactions grow increasingly common in VR environments, play will become an integral part of these interactions. This work presents an exploratory study on how different levels of customizability can enable increased creativity and interaction between players in a social VR environment.

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